

Question ID: 7a6d06bf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A rectangle has an area of **155** square inches. The length of the rectangle is **4** inches less than **7** times the width of the rectangle. What is the width of the rectangle, in inches?

Correct Answer: 5

Rationale

The correct answer is **5**. Let x represent the width, in inches, of the rectangle. It's given that the length of the rectangle is **4** inches less than **7** times its width, or $7x - 4$ inches. The area of a rectangle is equal to its width multiplied by its length. Multiplying the width, x inches, by the length, $7x - 4$ inches, yields $x(7x - 4)$ square inches. It's given that the rectangle has an area of **155** square inches, so it follows that $x(7x - 4) = 155$, or $7x^2 - 4x = 155$. Subtracting **155** from both sides of this equation yields $7x^2 - 4x - 155 = 0$. Factoring the left-hand side of this equation yields $(7x + 31)(x - 5) = 0$. Applying the zero product property to this equation yields two solutions: $x = -\frac{31}{7}$ and $x = 5$. Since x is the rectangle's width, in inches, which must be positive, the value of x is **5**. Therefore, the width of the rectangle, in inches, is **5**.